

Figure 6-22. Cutaway view of automotive hydraulic shock absorber. *(Courtesy of Texaco Inc., New York, New York.)*

6.10 KEY EQUATIONS

Cylinder extension force:

$$F_{ext} = pA_p \tag{6-1}$$

Cylinder extension velocity:

$$v_{ext} = \frac{Q_{in}}{A_p} \tag{6-2}$$

Cylinder retraction force:

$$F_{ret} = p\,(A_p - A_r) \tag{6-3}$$

Cylinder retraction velocity:

$$v_{ret} = \frac{Q_{in}}{A_p - A_r} \tag{6-4}$$

Cylinder power for extension and retraction strokes:

English units:

$$\text{HP} = \frac{v\,(\text{ft/s}) \times F(\text{lb})}{550} = \frac{Q_{in}(\text{gpm}) \times p(\text{psi})}{1714} \tag{6-5}$$

Metric units:

$$\text{kW power} = v\,(\text{m/s}) \times F(\text{kN}) = Q_{in}(\text{m}^3/\text{s}) \times p(\text{kPa}) \tag{6-5M}$$

First-class lever:

$$F_{\text{cyl}} = \frac{L_2}{L_1 \cos \phi} F_{\text{load}} \tag{6-7}$$

Second-class lever:

$$F_{\text{cyl}} = \frac{L_2}{(L_1 + L_2)\cos \phi} F_{\text{load}} \tag{6-8}$$

Third-class lever:

$$F_{\text{cyl}} = \frac{L_1 + L_2}{L_2 \cos \phi} F_{\text{load}} \tag{6-9}$$

Cylinder piston deceleration due to cushion:

$$a = \frac{v^2}{2S} \tag{6-10}$$

Maximum cylinder pressure developed by cushion:

$$p_2 = \frac{(\text{W}/g)a + p_1(A_{\text{piston}}) - (\text{CF})\text{W}}{A_{\text{piston}} - A_{\text{cushion plunger}}} \tag{6-11}$$

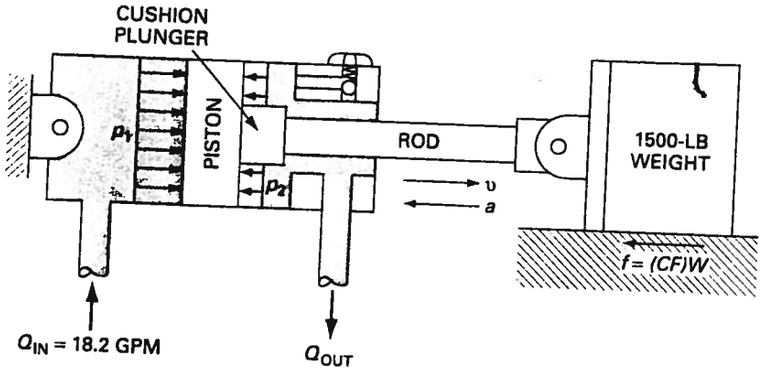


Figure 6-17. Cylinder cushion problem for Example 6-6.

Also note that mass m equals weight W divided by the acceleration of gravity g. The frictional retarding force f between the weight W and its horizontal support surface equals CF times W. This frictional force is the external load force acting on the cylinder while it is moving the weight.

Substituting into Newton's equation yields

$$p_2(A_{\text{piston}} - A_{\text{cushion plunger}}) + (CF)W - p_1(A_{\text{piston}}) = \frac{W}{g}a$$

Solving for p_2 yields a usable equation:

$$p_2 = \frac{(W/g)a + p_1(A_{\text{piston}}) - (CF)W}{A_{\text{piston}} - A_{\text{cushion plunger}}} \tag{6-11}$$

Substituting known values produces the desired result:

$$p_2 = \frac{[(1500)(5.51)/32.2] + 750(\pi/4)(3)^2 - (0.12)(1500)}{(\pi/4)(3)^2 - (\pi/4)(1)^2}$$

$$p_2 = \frac{257 + 5303 - 180}{7.07 - 0.785} = \frac{5380}{6.285} = 856 \text{ psi}$$

Thus the hydraulic cylinder must be designed to withstand an operating pressure of 856 psi rather than the pressure relief valve setting of 750 psi.

Example 6-6

A pump delivers oil at a rate of 18.2 gpm into the blank end of the 3-in-diameter hydraulic cylinder shown in Figure 6-17. The piston contains a 1-in-diameter cushion plunger that is 0.75 in long, and therefore the piston decelerates over a distance of 0.75 in at the end of the extension stroke. The cylinder drives a 1500-lb weight, which slides on a flat horizontal surface having a coefficient of friction (CF) equal to 0.12. The pressure relief valve setting equals 750 psi. Therefore, the maximum pressure (p_1) at the blank end of the cylinder equals 750 psi while the cushion is decelerating the piston.

Solution

Step 1: Calculate the steady-state piston velocity v prior to deceleration:

$$v = \frac{Q_{\text{pump}}}{A_{\text{piston}}} = \frac{(18.2/449)\text{ ft}^3/\text{s}}{[(\pi/4)(3)^2/144]\text{ ft}^2} = \frac{0.0406}{0.049} = 0.83 \text{ ft/s}$$

Step 2: Calculate the deceleration a of the piston during the 0.75-in displacement S using the constant acceleration (or deceleration) equation:

$$v^2 = 2aS$$

Solving for deceleration, we have

$$a = \frac{v^2}{2S} \tag{6-10}$$

Substituting known values, we obtain the value of deceleration:

$$a = \frac{(0.83 \text{ ft/s})^2}{2(0.75/12 \text{ ft})} = 5.51 \text{ ft/s}^2$$

Step 3: Use Newton's law of motion: The sum of all external forces ΣF acting on a mass m equals the product of the mass m and its acceleration or deceleration a:

$$\Sigma F = ma$$

When substituting into Newton's equation, we shall consider forces that tend to slow down the piston as being positive forces. Also the mass m equals the mass of all the moving members (piston, rod, and load). Since the weight of the piston and rod is small

Problems

Note: The letter *E* following an exercise number means that English units are used. Similarly, the letter *M* indicates metric units.

Hydraulic Cylinder Force, Velocity, and Power

6-14M. An 8-cm-diameter hydraulic cylinder has a 4-cm-diameter rod. If the cylinder receives flow at 100 Lpm and 12 MPa, find the

- **a.** Extension and retraction speeds
- **b.** Extension and retraction load carrying capacities

6-15E. A 3-in-diameter hydraulic cylinder has a 1.5-in-diameter rod. Find the flow-rate leaving the cylinder when it is extending by an entering flow-rate of 8 gpm.

6-16E. A pump supplies oil at 25 gpm to a $1\frac{1}{2}$-in-diameter double-acting hydraulic cylinder. If the load is 1200 lb (extending and retracting) and the rod diameter is $\frac{3}{4}$ in, find the

- **a.** Hydraulic pressure during the extending stroke
- **b.** Piston velocity during the extending stroke
- **c.** Cylinder horsepower during the extending stroke
- **d.** Hydraulic pressure during the retracting stroke
- **e.** Piston velocity during the retracting stroke
- **f.** Cylinder horsepower during the retracting stroke

Practice Problems.

Note: Add rod diameter selection to some of the problems by adding "extension = 12" or 16 inches"

Chapter 6

Solutions

6-15. 6.00 gpm

6-17. **(a)** 3.98 MPa, **(b)** 1.27 m/s, **(c)** 6.37 kW, **(d)** 5.31 MPa, **(e)** 1.70 m/s, **(f)** 8.50 kW

6-19. $1/4 \times p \times A_{piston}$

6-21. 354 psi

6-23. 2.19 in

6-25. **(a)** 4444 N, 2222 N, 8888 N, **(b)** 4444 N, 2222 N, 8888 N

6-27. 8480 lb

6-17M. A pump supplies oil at 0.0016 m^3/s to a 40-mm-diameter double-acting hydraulic cylinder. If the load is 5000 N (extending and retracting) and the rod diameter is 20 mm, find the

- **a.** Hydraulic pressure during the extending stroke
- **b.** Piston velocity during the extending stroke
- **c.** Cylinder kW power during the extending stroke
- **d.** Hydraulic pressure during the retracting stroke
- **e.** Piston velocity during the retracting stroke
- **f.** Cylinder kW power during the retracting stroke

6-18. Determine the value of constants C_1 and C_2 in the following equations for determining the speed of a hydraulic cylinder.

$$v\ (\text{in/min}) = \frac{C_1 Q(\text{gpm})}{A(\text{in}^2)} \qquad v\ (\text{m/s}) = \frac{C_2 Q(\text{m}^3/\text{s})}{A(\text{m}^2)}$$

6-19. A hydraulic cylinder has a rod diameter equal to one-half the piston diameter. Determine the difference in load-carry capacity between extension and retraction if the pressure is constant.

6-20. For the cylinder in Exercise 6-19, what would happen if the pressure were applied to both sides of the cylinder at the same time?

Cylinder Loads Due to Moving of Weights

6-21E. The inclined cylinder of Figure 6-23 has a 3-in-diameter piston. Determine the pressure required to extend the 5000-lb weight.

6-22M. A 10,000 N weight is to be lowered by a vertical cylinder as shown in Figure 6-24. The cylinder has a 75-mm-diameter piston and a 50-mm-diameter rod. The weight is to decelerate from 100 m/min to a stop in 0.5 s. Determine the required pressure in the rod end of the cylinder during the deceleration motion.

6-23E. A 6000-lb weight is being pushed up an inclined surface at constant speed by a cylinder as shown in Figure 6-25. The coefficient of friction between the weight and inclined surface equals 0.15. For a pressure of 1000 psi, determine the required cylinder piston diameter.

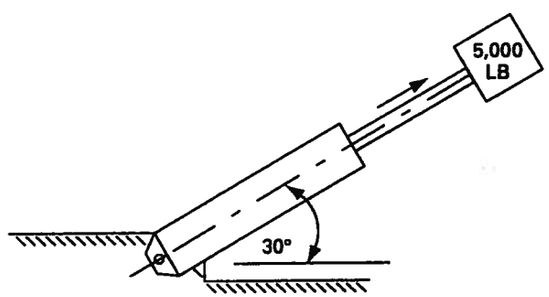


Figure 6-23. System for Exercise 6-21.

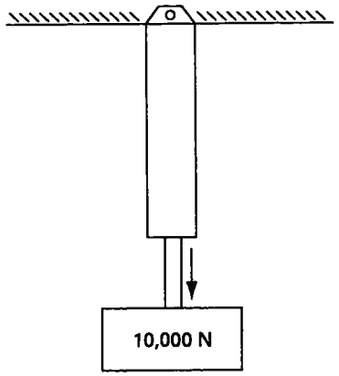


Figure 6-24. System for Exercise 6-22.

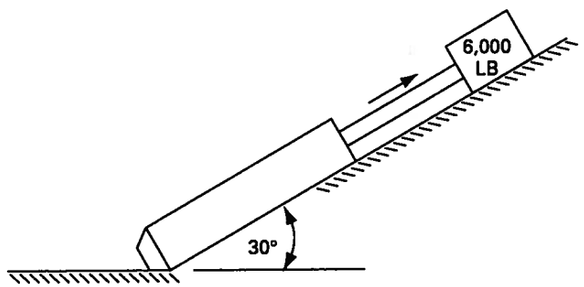


Figure 6-25. System for Exercises 6-23 and 6-24.

6-24E. Solve the problem of Exercise 6-23 if the weight is to accelerate from a zero velocity to a velocity of 5 ft/s in 0.5 s.

Cylinder Loadings Through Mechanical Linkages

6-25M. Change the data of Example 6-5 to metric units.

a. Solve for the cylinder force required to overcome the load force for the first-, second-, and third-class lever systems.

b. Repeat part a with $\theta = 10°$.

c. Repeat part a with $\phi = 5°$ and 20°.

6-26M. For the system of Figure 6-26, determine the hydraulic cylinder force required to drive a 1000-N load.

6-27E. For the crane system of Figure 6-27, determine the hydraulic cylinder force required to lift the 2000-lb load.

6-28E. For the toggle mechanism of Figure 6-28, determine the output load force for a hy-

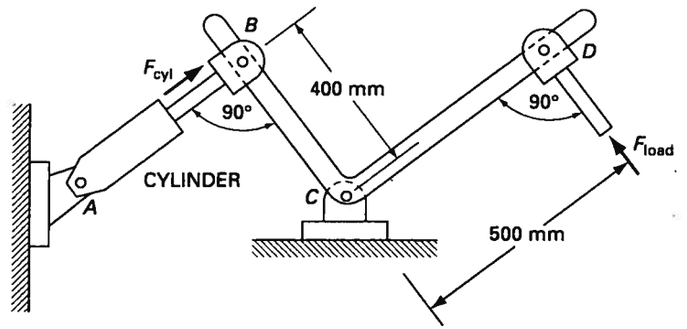


Figure 6-26. System for Exercise 6-26.

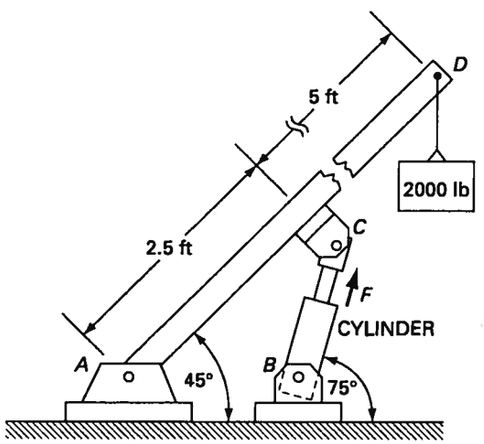


Figure 6-27. System for Exercise 6-27.

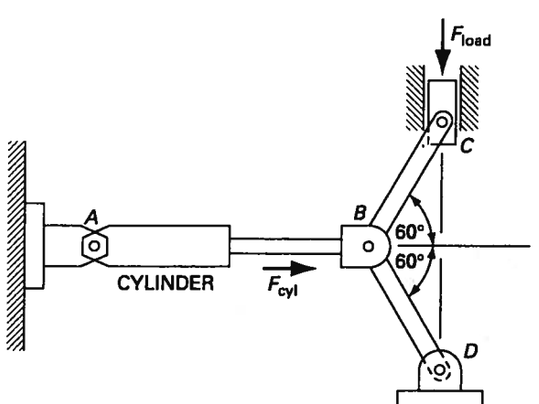


Figure 6-28. System for